SAHIL SURAJ JIRAPURE

B.Tech. - Information Technology

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PROFESSIONAL SUMMARY

Motivated and detail-oriented B.Tech student with a strong interest in software development, backend engineering, web development, and AI-driven solutions. Skilled in Java, MySQL, HTML, CSS, JavaScript, and problem-solving using Data Structures and Algorithms. Experienced in building academic and personal projects such as Gym Management System, College Resource Sharing Platform, Queue Management System, and Digital Marketing websites. Passionate about learning modern technologies, improving development skills, and contributing to innovative projects that create real-world impact. Seeking opportunities to gain practical experience, grow as a software developer, and contribute effectively to organizational success.

KEY EXPERTISE

Java | Python | HTML | CSS | JavaScript | MySQL | MongoDB | Data Structures and Algorithms | Object-Oriented Programming | Web Development | Backend Development | Database Management System | Operating Systems | GitHub | Git | VS Code | Problem Solving | Responsive Web Design | AI Tools | Automation | REST API Basics | Team Collaboration

EDUCATION

MIT School of Computing — 2023 - 2027

B.Tech. - Information Technology | CGPA: **7.65** / **10**

12th | Percentage: **62.80** / **100** — 2023

10th | Percentage: **69** / **100** — 2021

PROJECTS

AI-Powered Employee Attrition Prediction System
12 Jan, 2026 - 28 Apr, 2026

Mentor: College Project | **Team Size:** 3

Key Skills:

Python | Pandas | XGBoost | Machine Learning | Data Visualization | Predictive Analytics | Feature Engineering | HR Analytics | Model Evaluation

Project Link: https://github.com/sahil7751

Developed an AI-powered employee attrition prediction system designed to help organizations identify employees at high risk of attrition and enable proactive retention strategies. Built predictive machine learning models using Python and XGBoost to analyze workforce patterns, employee behavior, tenure, department trends, and performance-related factors. Implemented data preprocessing, feature engineering, and model evaluation pipelines to improve prediction accuracy and interpretability. Developed attrition risk scoring mechanisms along with feature importance analysis to provide actionable business insights for HR teams and management. Integrated cohort-level analytics and dashboard-ready reporting workflows for department-wise and tenure-based analysis using data visualization techniques. Contributed across data analysis, machine learning pipeline development, model optimization, and insight generation. The project demonstrated practical expertise in predictive analytics, applied machine learning, business intelligence, and data-driven decision-making for enterprise HR solutions.

Fake News Detection System
11 Nov, 2025 - 25 Feb, 2026

Mentor: Self | **Team Size:** 1

Key Skills:

Python | Scikit-learn | Natural Language Processing (NLP) | Machine Learning | Data Preprocessing | Feature Engineering | Model Evaluation | Classification Algorithms | Data Analysis

Project Link: https://github.com/sahil7751/Fake-News-Detection-System

Developed an AI-powered fake news detection system designed to classify news content into REAL, FAKE, or UNCERTAIN categories using machine learning and natural language processing techniques. Built a complete text preprocessing and feature extraction pipeline to improve classification accuracy and model reliability across diverse news datasets. Implemented and compared multiple machine learning models using evaluation metrics such as precision, recall, F1-score, and accuracy to identify the most effective approach for misinformation detection. Designed a prediction-ready workflow capable of processing user-provided news content and generating confidence-based classification outputs. Contributed to data cleaning, NLP pipeline development, feature engineering, model training, evaluation, and system optimization using Python and Scikit-learn. The project demonstrated practical expertise in machine learning, NLP, analytical problem-solving, and AI-driven content classification systems applicable to real-world media verification workflows.

bit-chats (Decentralized Messaging App) 20 Aug, 2025 - 25 Nov, 2025

Mentor: Prof. Rupali Bhatkhande | **Team Size:** 2

Key Skills:

rust | Bluetooth Mesh Networking | Torts | Peer-to-Peer Systems | End-to-End Encryption | Distributed Systems | Secure Communication | Network Programming | System Design

Project Link: https://github.com/sahil7751/Bit-chats

Designed and developed a decentralized messaging platform focused on enabling secure and private communication without relying on centralized servers or constant internet connectivity. Built an offline-first peer-to-peer communication system using Bluetooth mesh networking for message propagation across nearby devices. Implemented end-to-end encrypted messaging pipelines with secure key handling to ensure user privacy and data protection. Integrated Tor (Arti) support to enhance anonymity and secure routing capabilities within the communication layer. Contributed to system architecture, networking logic, encryption workflows, and decentralized message relay mechanisms using Rust. The project demonstrated strong understanding of distributed systems, secure communication protocols, low-level programming, and scalable decentralized application development.

FitSync - Fitness & Diet Recommendation Platform 15 Jan, 2025 - 20 Apr, 2025

Mentor: Self | **Team Size:** 1

Key Skills:

React | Node.js | Express.js | MongoDB | JavaScript | REST APIs | Authentication | Data Visualization | Responsive Web Design

Project Link: https://github.com/sahil7751/FitSync

Designed and developed a full-stack fitness and diet recommendation platform focused on improving user adherence to workout and nutrition plans through personalized recommendations and progress tracking. Built scalable REST APIs and responsive frontend interfaces using React, Node.js, Express.js, and MongoDB. Implemented features including personalized workout and calorie recommendations, nutrition logging with macro-level analytics, goal tracking, secure authentication, and mobile-optimized dashboards. Contributed across the complete development lifecycle including system design, backend architecture, API integration, database management, and UI development. The platform demonstrated strong problem-solving, full-stack engineering, and product development capabilities while showcasing practical experience in building user-centric health-tech solutions.

EduStake - College Resource Sharing Platform 10 Aug, 2024 - 25 Nov, 2024

Mentor: College Project | **Team Size:** 2

Key Skills:

JavaScript | Node.js | Express.js | MongoDB | REST APIs | Role-Based Access Control | Database Design | Backend Development | Responsive Web Development

Project Link: https://github.com/sahil7751

Designed and developed a centralized academic resource-sharing platform aimed at improving access to study materials, notes, assignments, and reference content for college students. Built a scalable full-stack application using Node.js, Express.js, MongoDB, and JavaScript with structured resource organization based on courses, semesters, and topics for efficient content discovery. Implemented secure role-based access control and contributor moderation workflows to maintain content quality and platform reliability. Developed upload, approval, and resource management functionalities along with usage analytics to identify high-value academic content and improve student engagement. Contributed across backend development, database schema design, API integration, authentication workflows, and responsive user interface development. The project demonstrated strong capabilities in full-stack engineering, backend architecture, database management, and building scalable educational technology solutions focused on usability and collaboration.

PUBLICATIONS / RESEARCH / WHITE PAPERS

Employee Attrition Prediction System using Machine Learning and Artificial Intelligence 27 Apr, 2026

Research Paper | IJCA | **Mentor:** Self | **No. of Authors:** 4

Key Skills:

Machine Learning | Artificial Intelligence | Python | XGBoost | Predictive Analytics | Data Analysis | Feature Engineering | HR Analytics | Model Evaluation | Data Visualization

Published a research paper focused on predicting employee attrition using machine learning and artificial intelligence techniques to support proactive workforce retention strategies. The research addressed the challenge of delayed identification of employee churn patterns in organizations, where traditional HR approaches often fail to detect attrition risks at an early stage. Proposed a predictive analytics framework capable of analyzing employee-related factors such as job satisfaction, tenure, workload, salary trends, department performance, and work environment indicators to estimate attrition probability.

The study involved data preprocessing, feature engineering, comparative model evaluation, and implementation of machine learning algorithms including XGBoost for high-accuracy prediction and interpretable results. Performance evaluation was conducted using metrics such as accuracy, precision, recall, and F1-score to validate model effectiveness. The proposed solution demonstrated how AI-driven HR analytics can assist organizations in reducing employee turnover, improving retention planning, and enabling data-driven decision-making. The research contributed practical insights into the application of machine learning for enterprise workforce analytics and intelligent business solutions.

ACHIEVEMENTS

- Built and deployed a personal portfolio website.
- Improved understanding of backend development and database design through practical projects.
- Continuously learning modern web technologies and AI-based tools for development.
- Developed multiple academic and personal software projects including Gym Management System and College Resource Sharing Platform.

CO-CURRICULAR ACTIVITIES

- Worked on collaborative software development projects in college.
- Participated in coding and technical project presentations.
- Engaged in learning sessions related to web development, databases, and AI technologies.

EXTRA CURRICULAR ACTIVITIES

- Exploring AI tools and automation technologies.
- Learning new development frameworks and backend technologies.
- Maintaining and improving personal development projects and portfolio.

PERSONAL INTERESTS / HOBBIES

- Coding
- Reading about Emerging Technologies
- Fitness and Health Improvement
- Learning AI Tools
- Reading Self Improving Books

WEB LINKS

- Github - https://github.com/sahil7751
- Personal - https://personal-portfolio-three-psi-59.vercel.app/

PERSONAL DETAILS

Gender: Male
Marital Status: Single
Current Address: MIT corner, Lonikalbhor, Pune, Pune, Maharashtra, India - 412201

Date of Birth: 31 Aug, 2005
Known Languages: English (Professional Proficiency), Hindi (Fluent), Marathi (Native),German(Basic) , Java (Intermediate), Python (Basic), SQL (Intermediate)
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